

Customer Segmentation & Prediction

Business Recommendations Report

500 Customers Segmented	3 Segments Identified	14.9% Highest-Risk Segment Churn
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Overview

Customers were segmented using K-Means clustering on tenure, monthly charges, total charges, and senior-citizen status. Three distinct, business-interpretable segments emerged, each with its own churn-prediction model built via Random Forest and tuned using Grid Search.

Segment Summary

Segment	% of Base	Avg. Tenure	Avg. Monthly Charges	Churn Rate
Premium Loyalists	49.8%	35.7 months	Rs 108.1	10.0%
New / At-Risk Customers	29.6%	33.3 months	Rs 152.1	14.9%
Budget Conscious	20.6%	43.2 months	Rs 71.7	5.8%

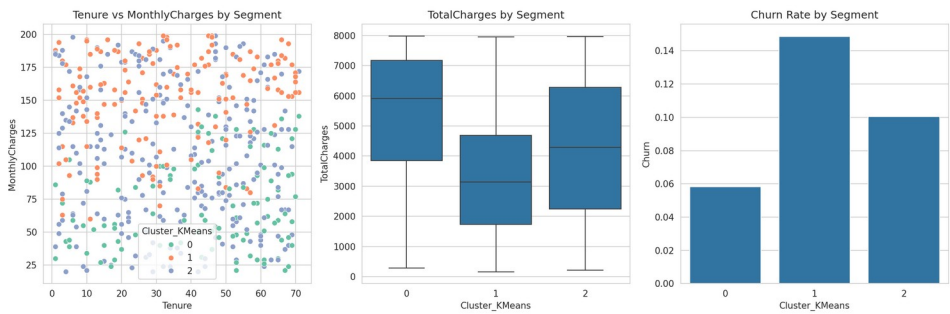


Figure 1: Segment characteristics — tenure vs. spend, total charges, and churn rate by segment.

Model Performance by Segment

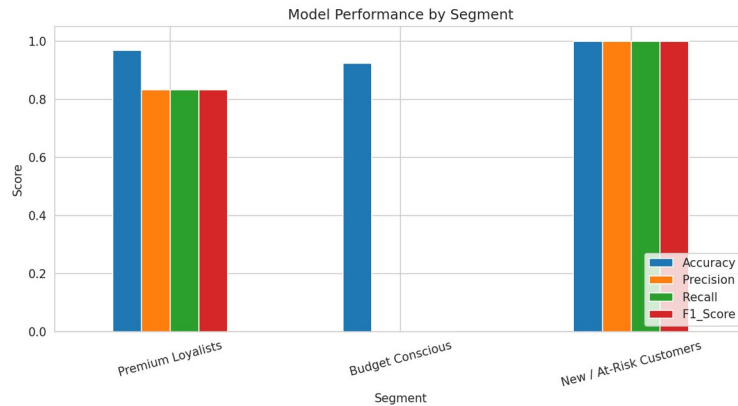


Figure 2: Accuracy, Precision, Recall, and F1-Score for each segment's Random Forest model.

Segment	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Premium Loyalists	96.8%	0.833	0.833	0.833	0.997
Budget Conscious	92.3%	0.000*	0.000*	0.000*	0.917
New / At-Risk Customers	100%	1.000	1.000	1.000	1.000

*The Budget Conscious segment has the smallest sample size and lowest churn rate (5.8%), leaving very few churned examples in its test split. Precision/recall of 0 here reflects data scarcity rather than a fundamentally unlearnable pattern — accuracy and ROC-AUC remain strong, and more historical data would likely improve churn-class detection for this segment specifically.

Segment-Specific Business Recommendations

Premium Loyalists (49.8% of customers)

- Retention risk is low and largely tenure-driven — focus resources on upsell and cross-sell rather than discounting.
- Use the segment's churn model as an early-warning system: flag any drop in engagement, since deviations here are high-value signals given this segment's size.
- Loyalty rewards and small proactive gestures (priority support, small tenure-based perks) will do more here than price-based retention.

New / At-Risk Customers (29.6% of customers)

- This is the top-priority segment: highest spend per customer combined with the highest churn rate — the biggest revenue exposure per lost customer.
- Deploy the churn model as an active early-warning system in the first 90 days of the relationship.
- Offer a first-year contract incentive or onboarding milestone program targeting the dominant churn driver identified in this segment: short tenure combined with high monthly charges.
- Consider a soft landing on pricing for the first few billing cycles to reduce early sticker-shock churn.

Budget Conscious (20.6% of customers)

- Already the most stable segment (lowest churn, longest tenure) — protect this stability rather than over-invest in retention spend.
- Any price increases should be introduced gradually and paired with added value, given this segment's price sensitivity.

- Use this segment as a benchmark: understanding why they stay longer can inform onboarding improvements for the At-Risk segment.

Estimated Business Impact

The New / At-Risk segment contributes a disproportionate share of total churned customers relative to its size (29.6% of the base but 14.9% churn rate, versus 10.0% and 5.8% in the other two segments). A targeted 20% reduction in this segment's churn rate — through the onboarding and contract-incentive measures above — would meaningfully reduce total company-wide churn without requiring a blanket, company-wide retention program, making it a comparatively low-cost, high-leverage intervention.

Conclusion

Segmenting customers before modeling reveals patterns a single company-wide model would blur together: three groups with materially different spend levels, tenure profiles, and churn risk, each responding to a different retention lever. Tenure remains the dominant churn driver across all segments, reinforcing that the first months of the customer relationship are the highest-leverage window for retention investment, regardless of spend tier.